

Electric Winch

Technical Datasheet - Generic Theatre / Entertainment Use

Document owner	Stage Electrics / Portal editor	Document type	Generic technical reference
Equipment category	Powered lifting machine	Portal item ID	electric_winch
Primary context	LOLER / PUWER / BS EN 17206 / ABTT	Status	Template for site-specific completion
Document title	Electric Winch Technical Datasheet	Revision	Draft for review

Important use note

- This is a generic technical datasheet for portal reference and inspection planning.
- It does not replace the manufacturer manual, installation drawings, commissioning records, load calculations or a competent person assessment.
- Complete the site-specific fields before relying on it for a particular installation.

1. Equipment overview

An electric winch is a powered lifting or pulling appliance used to raise, lower or move loads by rope, wire rope, belt or drum system. In theatre and entertainment environments, electric winches may be used for bars, scenery, curtains, projector screens, acoustic banners, masking systems or integrated lifting systems.

The winch should be assessed as part of a complete system, not as a stand-alone motor. The complete load path includes the winch body, drum, mounting frame, motor, gearbox, brake, rope or belt, terminations, pulleys, diverters, guides, control system, limits, support structure and the suspended or moved load.

2. Site-specific technical details to complete

Field	Information to record	Field	Information to record
Manufacturer	To be completed	Model / type	To be completed
Serial number / asset ID	To be completed	Location / venue	To be completed
SWL / WLL	To be completed	Duty / speed / travel	To be completed
Rope / belt type and diameter	To be completed	Number of falls / reeving	To be completed
Mounting method	To be completed	Support structure	To be completed
Control type	To be completed	Limit devices fitted	To be completed
Brake type	To be completed	Emergency stop / isolation	To be completed
Date installed	To be completed	Installed by	To be completed
Inspection interval	To be completed	Next examination due	To be completed

Field	Information to record	Field	Information to record
Relevant manuals / drawings	To be completed	Site limitations	To be completed

3. Principal components

Component	Purpose	Typical checks
Winch body / frame	Supports and houses the winch mechanism.	Secure mounting, no cracking, distortion, corrosion or movement.
Motor and gearbox	Provides powered movement and torque control.	No abnormal noise, overheating, oil leaks, looseness or damaged covers.
Brake / holding device	Holds the load and prevents unintended movement.	Holds load correctly; no slipping, drift or delayed response.
Drum / rope anchor	Stores and anchors the rope, wire rope or belt.	Correct winding, no crushing, bunching, damage or anchor movement.
Wire rope / rope / belt	Transfers force from the winch to the load.	No broken wires, kinks, crushing, wear, contamination, cuts or heat damage.
Pulleys / diverters / guides	Guide the rope or belt along the load path.	Free rotation, correct alignment, secure fixings and no sharp contact points.
Terminations	Connect the rope or belt to the load or anchor point.	Correct fitting, no slippage, deformation, broken wires, corrosion or missing parts.
Controls / limits / E-stop	Control movement and restrict travel.	Correct direction, working limits, emergency stop, isolation and labels.

4. Regulatory and good-practice context

Reference	How it relates to an electric winch	Practical portal wording
LOLER 1998	An electric winch used for lifting or lowering loads is lifting equipment. LOLER principles include strength, stability, marking, planning of lifting operations, and thorough examination by a competent person.	Treat as lifting equipment. Confirm WLL/SWL, load path, planning, examination reports and defect actions.
PUWER 1998	An electric winch is work equipment. It must be suitable, maintained in a safe condition, inspected where deterioration could create risk, and used by trained people.	Check suitability, maintenance records, user instructions, isolation, guarding, inspection and training.
BS EN 17206	Where the winch forms part of entertainment machinery for stages or production areas, this standard supports the assessment of load-bearing components, safety functions, control systems and prevention of uncontrolled movement.	Consider controls, brakes, limits, documentation, testing, commissioning and system integration.
ABTT Technical Standards	Provides recognised theatre-industry good-practice context for places of entertainment.	Use as good-practice support for installation, inspection, maintenance, records and venue-specific risk.

5. Installation and use considerations

- Confirm that the winch, support frame and fixings are suitable for the applied load, duty cycle, travel and operating environment.
- Check that the supporting structure has been assessed for static, dynamic and any foreseeable shock or side loading.
- Ensure the rope, belt or wire rope path is correctly aligned and free from sharp edges, crushing points or snagging points.
- Confirm that travel limits, emergency stops, braking devices and control functions are appropriate for the risk.
- Where the winch is part of a larger machinery system, check the system as a whole rather than only the winch unit.
- Do not alter control logic, limits, reeving, rope type or load path without competent assessment and suitable records.

Theatre-specific risk reminders

- Loads may be overhead, above performers, technicians or audience areas.
- Winches may be used intermittently, so deterioration or poor winding may be missed without regular checks.
- Temporary scenic changes can alter the load, balance, centre of gravity or operating risk.
- Access for inspection and maintenance should be considered during design and operation.

6. Pre-use and routine inspection checks

Area	Check	Accept / reject guide
Identification	WLL/SWL, asset ID, manufacturer plate and direction labels are present and legible.	Do not use if rating or identity is uncertain.
Mounting	Winch body, mounting bolts, frame and support structure are secure.	No movement, loose fixings, cracking, distortion or corrosion.

Area	Check	Accept / reject guide
Rope / belt	Inspect for wear, broken wires, kinks, crushing, cuts, contamination, corrosion or poor winding.	Remove from service if damage or deterioration is found.
Drum winding	Check rope or belt lays correctly on the drum without riding, crossing, bunching or crushing.	Stop use if winding is unsafe or uncontrolled.
Terminations	Check sockets, clamps, thimbles, anchors and load attachments.	Correctly assembled, secure, undamaged and traceable.
Controls	Check direction controls, speed control, emergency stop, isolation and warning indicators.	Controls operate correctly and match labels.
Limits	Check upper/lower or travel limits where safe and appropriate.	Limits stop travel before hazardous overrun.
Brake / holding	Check for load drift, slipping, delayed stopping or uncontrolled movement.	No drift or slipping. Remove from service if uncertain.
Operation	Listen and observe during movement.	No abnormal noise, vibration, jerking, overheating or uneven travel.
Records	Check inspection, maintenance and thorough examination status.	Do not continue where overdue or serious defects remain open.

7. Common defects and likely actions

Defect / concern	Why it matters	Typical action
Missing or unreadable WLL/SWL	The safe capacity cannot be confirmed.	Quarantine or verify rating before use.
Damaged wire rope / rope / belt	May reduce capacity and increase risk of failure.	Remove from service and refer to competent person.
Poor drum winding	Can crush the rope, cause uneven movement or overload parts of the system.	Stop use, unload safely and correct under competent supervision.
Brake slip or load drift	Indicates loss of load control.	Remove from service immediately.
Limit switch failure	Can allow over-travel and collision or overload.	Isolate until repaired and tested.
Loose mounting or frame movement	Can compromise the entire load path.	Remove from service and inspect supports/fixings.
Abnormal noise or vibration	May indicate gearbox, bearing, brake or alignment issue.	Stop and investigate before further use.
Unauthorised modification	May invalidate design assumptions, rating or safety functions.	Do not use until assessed and documented.

8. Documentation and records

- Manufacturer manual, installation instructions and maintenance instructions.
- Declaration of Conformity or equivalent product documentation where available.
- Installation drawings, load path information and support-structure assessment.
- Commissioning, functional test and limit-setting records.
- Thorough examination reports and defect close-out records.
- Planned maintenance history, inspection check sheets and repair records.
- User instructions, safe system of work and training/competence records.

9. Portal code reference

Upload the PDF version of this document to the GitHub documents folder using the filename below, then add the document reference to the electric_winch equipment item when the portal JavaScript is updated.

```
documents/electric_winch_technical_datasheet.pdf

documents: [
  {
    title: "Electric Winch Technical Datasheet",
    file: "documents/electric_winch_technical_datasheet.pdf"
  }
]
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10. Disclaimer

This datasheet is guidance only and is intended to support the Mechanical Theatre Equipment Compliance Portal. It does not certify the equipment, prove suitability, replace a competent person inspection, or override manufacturer instructions, statutory requirements, venue procedures, risk assessments or a written scheme of examination.